Assignment 2: Coding Basics

Abrão Soares Pereira

OVERVIEW

This exercise accompanies the lessons/labs in Environmental Data Analytics on coding basics.

Directions

1. Rename this file `<FirstLast>_A02_CodingBasics.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change "Student Name" on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.
6. After Knitting, submit the completed exercise (PDF file) to Canvas.

Basics, Part 1

1. Generate a sequence of numbers from one to 55, increasing by fives. Assign this sequence a name.

2. Compute the mean and median of this sequence.

3. Ask R to determine whether the mean is greater than the median.

4. Insert comments in your code to describe what you are doing.

```
#1.
my_sequence <- seq(from = 1, to = 55, by = 5)

#2.
mean_value <- mean(my_sequence)
median_value <- median(my_sequence)

#3.
mean_greater <- mean_value > median_value
print(mean_greater)
```

```
## [1] FALSE
```

```
# First I need to give name to my function, which is my_sequence.
#Then assign value to it. After that run the sequence by asking R to give me a
#sequence 5 of from number 1 to 55. After that find the mean and median value
#from my assign sequence. Ask R to find which one is greater mean or median
```

Basics, Part 2

5. Create three vectors, each with four components, consisting of (a) student names, (b) test scores, and (c) whether they are on scholarship or not (TRUE or FALSE).
6. Label each vector with a comment on what type of vector it is.
7. Combine each of the vectors into a data frame. Assign the data frame an informative name.
8. Label the columns of your data frame with informative titles.

```r
# Vector of student names (character vector)
student_names <- c("Joao", "Adriana", "Luisa", "Luan")

# Vector of test scores (numeric vector) 90 above is pass to get a scholarship but below 90 is not
test_scores <- c(96, 89, 94, 86)

# Vector indicating whether on scholarship (logical vector)
on_scholarship <- c(TRUE, FALSE, TRUE, FALSE)

# 7. Combine all the information into a data frame
student_data <- data.frame(student_names, test_scores, on_scholarship)

# 8. Column with labels
colnames(student_data) <- c("Names", "Score", "Scholarship")
print(student_data)
```

```
##     Names Score Scholarship
## 1    Joao    96        TRUE
## 2 Adriana    89       FALSE
## 3   Luisa    94        TRUE
## 4    Luan    86       FALSE
```

9. QUESTION: How is this data frame different from a matrix?

   Answer: Because it can store different data type. Unlike Matrix, where all of the elements must be the same

10. Create a function with one input. In this function, use `if...else` to evaluate the value of the input: if it is greater than 50, print the word "Pass"; otherwise print the word "Fail".
11. Create a second function that does the exact same thing as the previous one but uses `ifelse()` instead if `if...else`.
12. Run both functions using the value 52.5 as the input
13. Run both functions using the **vector** of student test scores you created as the input. (Only one will work properly. . .)

```r
#10. Create a function using if...else
check_pass_fail_if_else <- function(score) {
  if (score > 50) {
    print("Pass")
  } else {
```

```
    print("Fail")
  }
}

#11. Create a function using ifelse()
check_pass_fail_ifelse <- function(score) {
  print(ifelse(score > 50, "Pass", "Fail"))
}

#12a. Run the first function with the value 52.5
check_pass_fail_if_else(52.5)
```

```
## [1] "Pass"
```

```
#12b. Run the second function with the value 52.5
check_pass_fail_ifelse(52.5)
```

```
## [1] "Pass"
```

```
#13a. Run the first function with the vector of test scores
# check_pass_fail_if_else(test_scores)

#13b. Run the second function with the vector of test scores
check_pass_fail_ifelse(test_scores)
```

```
## [1] "Pass" "Pass" "Pass" "Pass"
```

14. QUESTION: Which option of `if...else` vs. `ifelse` worked? Why? (Hint: search the web for "R vectorization")

    Answer: The ifelse() function worked correctly with the vector of test scores. if…else is only evaluates the very first element of a vector. While, ifelse() it efficiently performs the conditional check on every element of the input vector.

NOTE Before knitting, you'll need to comment out the call to the function in Q13 that does not work. (A document can't knit if the code it contains causes an error!)